

	ITER,SIKSHA‘O’ANUSANDHAN (Deemed to be University)		LESSONPLAN
Programme	B.Tech.	AcademicYear	2024-25
Department	CSE, CSIT	Semester	5 th
Credit	3	GradingPattern	6
SubjectCode	CSE 3731		
SubjectName	Introduction to the Theory of Computation		
WeeklyCourseFormat	3 Classes/week, 1 hr/Class; 3 Credits		
SubjectCoordinator(s)	Dr. Lambodar Jena Dr. Prithviraj Mohanty Dr. Nimisha Ghosh		
TextBooks(s): (1) Introduction to the Theory of Computation, by Michael Sipser, Cengage learning.			
CourseOutcomes	Studentswillbeable to		
	CO1	Enhance/develop ability to understand and conduct mathematical proofs for computation and algorithms.	
	CO2	Design and analyze finite automata and regular expression for describing regular languages.	
	CO3	Design and analyze pushdown automata, and context-free grammars	
	CO4	Design and analyze Turing machines	
	CO5	Enhance the ability to understand the decidability, undecidability, and reducibility criteria of various computational problems.	
	CO6	Demonstrate the understanding of key notions, such as algorithm, computability and complexity through problem solving.	

CSE3731:Introduction to the Theory of Computation

Sl.No.	Lessons/Topic to be covered	Book Reference (sections)	Mapping with COs	Home Work/Assignments/Quizzes
1	Introduce the grading pattern, credit, Evaluation strategy, Major contents of the course. Motivation behind the course. Introduction to automata, computability and complexity theory.	Sipser (pg.1-3)	CO1	
2	Mathematical Notions and Terminology: Set, Sequences and tuples, Tree, Graphs, Strings and Languages	Sipser (pg.3-14)	CO1	
3	Types of Proof: Proof by construction.	Sipser (pg.21)	CO1	
4	Types of Proof: Proof by contradiction, Proof by induction.	Sipser (pg.21-22)	CO1	
5	Regular Languages: Finite Automata, Formal definition of a finite automaton, Examples of finite automata	Sipser (pg.31-37)	CO2	
6	Designing finite automata.	Sipser (pg.41)	CO2	
7	Designing finite automata contd...	Sipser (pg.41)	CO2	
8	Regular operations (union, concatenation, star).	Sipser (pg.44)	CO2	
9	Nondeterminism: Formal definition of a nondeterministic finite automaton.	Sipser (pg.47-53)	CO2	Quiz-1
10	NFA examples and sample design.	Sipser (pg.54)	CO2	
11	Designing NFA contd...	Sipser (pg.54)	CO2	
12	Equivalence of NFAs and DFAs.	Sipser (pg.54)	CO2	
13	Introduction to regular expressions. Formal definition of Regular expression	Sipser (pg.63-64)	CO2	
14	Equivalence of regular expression and finite automata.	Sipser (pg.66)	CO2	
15	Non-regular languages: The pumping lemma for regular languages	Sipser (pg.77)	CO2	
16	Examples on pumping lemma.	Sipser (pg.78-82)	CO1, CO2	
17	Context-Free Languages: Context-Free Grammars, Formal Definition of a context-free grammar.	Sipser (pg.102-104)	CO3	Quiz-2
18	Examples of context-free grammars	Sipser (pg.105-106)	CO3	
19	Designing context-free grammars.	Sipser (pg.105-106)	CO3	
20	Ambiguity	Sipser (pg.107)	CO3	
21	Chomsky normal form (CNF).	Sipser (pg.107)	CO3	Assignment-1
22	Introduction to pushdown automata (PDA): formal definition of a pushdown automata	Sipser (pg.111-113)	CO3	
23	Examples on pushdown automata.	Sipser (pg.114)	CO3	
24	Equivalence with context-free grammars.	Sipser (pg.117)	CO3	

CSE3731: Introduction to the Theory of Computation

Sl.No.	Lessons/Topics to be covered	Book Reference (sections)	Mapping with COs	Home Work/Assignments/Quizzes
25	Non-context-free languages: The pumping lemma for context-free languages.	Sipser (pg.125)	CO3	
26	Examples on the pumping lemma for CFG	Sipser (pg.128-129)	CO3	Quiz-3
27	Deterministic context-free languages (DCFL): Properties of DCFLs	Sipser (pg.130-133)	CO3	
28	Deterministic context-free grammars	Sipser (pg.135)	CO3	
29	The Church-Turing Thesis, Turing Machines: Formal definition of a Turing machine.	Sipser (pg.165-167)	CO4	
30	Examples of Turing machines.	Sipser (pg.170)	CO4	Assignment-2
31	Variants of Turing machines, The Definition of Algorithm: Hilbert's Problem.	Sipser (pg.176-178, 182)	CO4, CO6	
32	Decidability: Decidable Languages, Decidable problems concerning regular languages	Sipser (pg.193-198)	CO5	
33	Decidable problems concerning context-free languages.	Sipser (pg.198)	CO5	
34	Undecidability: The diagonalization method. An undecidable language. A Turing-unrecognizable language.	Sipser (pg.201-209)	CO5	Quiz-4
35	Reducibility: undecidable problems from language theory.	Sipser (pg.215-216)	CO5, CO6	
36	A simple undecidable problem, mapping reducibility	Sipser (pg.227-234)	CO5, CO6	